

Voice AI — V3 Planning

Architecture decisions

- Twilio **subaccounts** per PM (Twilio's official multi-tenant pattern)
- Credentials **encrypted at rest** in database
- Onboarding via **unique invite link** (one-time use, no expiry)
- Payments **handled outside the system** in V3.0 — controls access via invite
- **State persistence** on provisioning to allow recovery after failure

V3.0 — MVP

Goal: a second PM can use the system independently, without touching code.

Estimated total: ~56 hrs / ~7 business days (1 dev)

#	Area	Task	Details	Est.
1	DB	voice_clients table	Stores each PM's configuration, credentials and account status	2h
2	DB	vc_invites table	Tracks invite links: generated, reserved, used	2h
3	Core	VoiceClientContext	Object that holds the active PM's config during a request	2h
4	Core	ResolveVoiceClientMiddleware	Identifies which PM is being called based on the dialed number	3h
5	Core	RentManagerHttpClient refactor	Makes the Rent Manager connection use each PM's own credentials	8h
6	Core	RentManagerAuth	Keeps each PM's	2h

#	Area	Task	Details	Est.
		hService	Rent Manager session token separate in cache	
7	Core	VoiceAIController	Ties each incoming call to the correct PM before the AI session starts	4h
8	Core	VoiceAIWebSocketsHandler	Ensures the AI session uses the correct PM's data throughout the call	4h
9	Core	VoiceAIService + TwilioClient	Uses each PM's own support phone and Twilio credentials	3h
10	Provisioning	TwilioProvisioningService	Creates the PM's Twilio subaccount, buys their number, registers the webhook. Saves progress at each step so nothing is lost if something fails	10h
11	UI	Admin — Generate Invite	Simple page for Admin: one button generates an invite link to share with the PM	4h
12	UI	Form — Step 1 (validation)	PM fills in their details; system validates Rent Manager credentials before spending	6h

#	Area	Task	Details	Est.
			anything	
13	UI	Form — Step 2 (confirmation)	PM reviews and confirms; system provisions their account and displays their assigned number	6h
14	Migration	Migrate existing config	Move current setup into the new multi-PM structure with no downtime	8h

Key risks: RentManagerHttpClient refactor (#5) touches the core of the system — requires full regression test on flow. Provisioning (#10) makes real Twilio API calls — must be tested in Twilio test mode first. Migration (#14) must be done with zero downtime in production.

V3.1 — Post-MVP

Goal: polished flow, self-service and operational visibility.

Estimated total: ~24–44 hrs / ~3–5.5 days (depending on payment decision)

#	Area	Task	Details	Est.	Depends on
1	Email	Email service setup	Configure a transactional email provider (e.g. SendGrid)	3h	Client decision
2	Email	Notification templates	"Account approved" and "Account active — your number is X" emails	5h	#1
3	Admin	voice_clients management panel	List of all PM accounts with statuses, retry failed	10h	—

#	Area	Task	Details	Est.	Depends on
			provisioning, suspend		
4	Onboarding	Public registration (no invite)	Anyone can register from a public page; replaces invite-only flow	6h	Client decision
5	Payments	Stripe integration	Automated payment before provisioning, with full webhook handling	20h	Client decision

Summary

Version	What's included	Estimate	Outcome
V3.0	Multi-tenant core + Twilio provisioning + invite-based onboarding	~7 business days	A second PM can operate independently
V3.1	Emails + admin panel + public registration + Stripe (optional)	~3–5.5 days	Polished product, self-service, full visibility
Total		~10–12.5 business days	

Client Validation Questions

Money

- Does SymAssist earn a margin on top of costs, or do PMs just pay the platform cost (Twilio + infra)?
- Is payment handled inside the system (checkout) or externally (invoice, bank transfer)?

- If provisioning fails after a payment is made — human intervention to fix it, or automatic retry with refund if unrecoverable?
- Who absorbs the Twilio number cost (~\$1/mo per PM)? Platform or PM directly?

UX

- Personal invite links or public registration page anyone can find?
- Do PMs need to see the status of their own account (pending, active, suspended)?
- How does the PM find out their account is ready and what their number is? (email, manual contact, shown on screen?)
- Should PMs be able to update their own Rent Manager credentials if they change, or is that an admin operation?

Strategy

- Twilio numbers managed under one platform account (simpler, recommended) or each PM gets their own Twilio account (full isolation)?
- What happens when a PM cancels or is suspended — is their number released, archived, or transferred?
- If a PM leaves, do they keep ownership of their Twilio number or does it stay with the platform?

Product

- Should the IVR greeting be customizable per PM (business name, welcome message) or is the same script fine for everyone?
- Is transfer-to-human a required feature from day one, or optional per PM?
- What information do we need from the PM at registration beyond Rent Manager credentials? (contact name, billing email, company name, human redirection number)
- Should the system support multiple languages per PM, or is English-only sufficient for now?

Operations

- How many PMs are expected in the first wave, and what's the growth target for year one?
- Is there a specific onboarding SLA? (e.g. "account must be active within X minutes of form submission")
- Does each PM's AI need to know their specific business name, or is a generic configuration acceptable?

ANSWERS - CLIENT VALIDATION QUESTIONS (Doug, Daily Sync 25 Jun 2026)

Captured from the call with Doug on 2026-06-25. Items marked PENDING were not covered yet; Doug will bring the Product section next week.

MONEY

- Margin or just cost pass-through? Doug: SymAssist earns a margin. The per-unit price must cover all operating costs PLUS profit.
- Payment inside the system or external? Doug: External, handled in Zoho (native

payments + automated bank reconciliation). Billing is SymAssist-to-PM directly; Rent Manager is not the merchant of record.

- Provisioning fails after payment - manual fix or auto retry/refund? PENDING (not discussed).
- Who absorbs the ~\$1/mo Twilio number cost? Doug: The platform absorbs it and recovers it through the monthly per-unit fee.

UX

- Personal invite links or public registration page? Doug: Neither - the PM clicks Request Activation in Rent Manager and then receives a SymAssist sign-up link. V3.0 stays request-based, no public page.
- Do PMs see their own account status (pending/active/suspended)? Doug: No.
- How does the PM learn the account is ready and their number? Doug: By email.
- Can PMs self-update their Rent Manager credentials? Doug: No - admin operation. With the RM partnership/partner token we likely won't need to collect the API key from the PM at all.

STRATEGY

- One platform Twilio account or one per PM? Doug: One platform account. PMs won't even know the word Twilio.
- On cancel/suspend - is the number released, archived or transferred? Doug: Solved via forwarding - we forward the PM's existing business number to the platform Twilio number, so canceling just means we stop forwarding (immediate).
- If a PM leaves, who owns the Twilio number? Doug: It stays with the platform. The PM keeps their own legacy number, which was only being forwarded.

PRODUCT

- IVR greeting customizable per PM? PENDING - Doug will confirm next week.
- Transfer-to-human required from day one? PENDING - Doug will confirm next week.
- What info do we need from the PM at registration beyond RM credentials? Mostly covered by the RM activation (contact name, phone and email come through); with the partner token the API key is not needed from the PM. No extra field list finalized.
- Multiple languages per PM or English-only? English only

OPERATIONS

- PMs in the first wave / year-one growth target? Doug: ~10 design partners for the first few months, then 1-2 per week. A significant share of the first ~50 clients will be onboarded for free.
- Onboarding SLA? Doug: No strict minutes SLA; target is the account ready within ~24h, excluding the phone-number forwarding step.
- Per-PM business name in the AI or generic config? PENDING (part of the skipped Product section).